

A Hybrid Machine Learning and Domain Reputation Framework for Phishing URL Detection

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Abstract - Phishing attacks remain one of the most prevalent cybersecurity threats, exploiting deceptive URLs and fraudulent domains to steal sensitive user information. Traditional phishing detection techniques, such as blacklist-based and rule-based systems, are largely reactive and ineffective against zero-day and short-lived phishing websites. To address these limitations, this paper proposes a hybrid phishing detection framework that integrates machine learning-based URL analysis with domain reputation assessment using DNS and SSL certificate indicators. The proposed approach employs an Extreme Gradient Boosting (XGBoost) classifier to analyze lexical and structural URL features, while trust-based indicators derived from DNS resolution and SSL certificate validation provide complementary contextual verification. A weighted hybrid decision strategy is adopted to combine both components, improving robustness and reducing false-positive rates. Experimental evaluation conducted on a balanced dataset of phishing and legitimate URLs demonstrates the effectiveness of the proposed framework, achieving an overall detection accuracy of 99% with high precision, recall, and F1-score values. The results further indicate strong generalization capability and excellent class separability, as evidenced by ROC and Precision-Recall analysis. The proposed framework offers a scalable, lightweight, and real-time phishing detection solution without reliance on proprietary or paid threat intelligence services, making it suitable for practical deployment in real-world cybersecurity environments.

Keywords- Phishing Detection, URL-Based Analysis, XGBoost, Machine Learning, DNS Reputation, SSL Certificate Analysis, Hybrid Detection Framework, Cybersecurity

1. INTRODUCTION

Phishing attacks have emerged as one of the most prevalent and damaging threats in the modern cyber security landscape, targeting users by impersonating legitimate websites and online services to steal sensitive information such as login credentials, financial data, and personal information [1]. The expansion of internet and online services to various services such as banking , e-commerce , gaming ,

social media etc has attracted various malicious attackers who try to steal critical data through phishing. These attackers continuously evolve phishing techniques that exploit user trust through deceptive URLs, fraudulent domains, and visually similar web interfaces. Studies have shown that phishing websites are often short-lived and dynamically generated, making timely and accurate detection particularly challenging [1].

Traditional phishing detection mechanisms, including blacklist-based and rule-based approaches, have been widely adopted in browsers and security tools [1], [4]. However, such approaches are inherently reactive in nature and largely ineffective against zero-day and newly created phishing websites [1]. The blacklist-based systems fail to identify unseen phishing attacks in early stages as they rely on previously reported phishing domains which leaves the user vulnerable. These limitations have motivated the shift toward intelligent and automated phishing detection techniques [1], [3].

Machine learning-based phishing detection has gained significant attention due to its ability to learn hidden patterns from data and generalize to previously unseen attacks [2], [3]. Especially, URL-based machine learning approaches leverage lexical and structural characteristics of URLs to distinguish phishing websites from legitimate ones with low computational overhead [2], [4]. These methods provide an effective solution for early-stage detection and real-time deployment. However, the performance of machine learning models is highly dependent on effective feature engineering and model selection, and single-model approaches may struggle to capture the diverse and evolving characteristics of phishing attacks [3], [8].

Deep learning-based approaches have shown high detection accuracy in phishing classification tasks but their practical deployment is often constrained by high computational complexity, large training data requirements, and increased latency [3], [9]. Due to which, lightweight ensemble learning techniques have gained preference for real-world phishing detection systems [10], [11]. Among these techniques, Extreme Gradient Boosting (XGBoost) has shown strong performance in handling non-linear feature

interactions, class imbalance, and feature importance analysis, making it a suitable choice for URL based phishing detection [10]–[12].

Some recent studies demonstrate that relying solely on URL-based machine learning is insufficient to address the complexity of modern phishing attacks [5], [6]. Hybrid phishing detection frameworks that incorporate additional trust-based indicators, such as SSL certificate properties and DNS related features, have been shown to improve detection robustness and reduce false-positive rates [5], [13], [14]. Trust-based heuristic indicators provide complementary contextual information that strengthens machine learning predictions, particularly for newly registered or suspicious domains.

Motivated by these observations, this work proposes a hybrid phishing detection framework that integrates URL-based machine learning using XGBoost with trust-based indicators obtained via secure web APIs. The proposed approach aims to achieve accurate, scalable, and real-time phishing detection while keeping in consideration the due practicality of such systems. The system seeks to provide a balanced and practical solution for phishing website detection by combining machine learning predictions with trust-based signals in dynamic online environments.

2. RELATED WORK

Phishing detection has been an active research area for over a decade, resulting in the development of multiple defensive techniques ranging from blacklist-based systems to intelligent machine learning-driven approaches [1]. Early survey research grouped the different phishing countermeasures into blacklist-based, heuristic-based, and machine learning-based approaches, and noted that all blacklist-based systems are inherently reactive and ineffective against zero-day and short-lived phishing Web sites [1]. These studies emphasized that no single detection technique is sufficient to address the diverse and evolving nature of phishing attacks, thus emphasizing the need for adaptive and intelligent detection frameworks [1], [3].

Machine learning-based phishing detection has gained prominence due to its ability to automatically learn patterns from data and generalize to previously unseen attacks [2], [3]. In particular, URL-based machine learning approaches leverage lexical and structural characteristics of URLs to distinguish phishing websites from legitimate ones while remaining lightweight and suitable for real-time deployment [2], [4]. Several of these approaches have been proven to outperform rule-based methods; however, they tend to be very dependent on feature engineering and model selection in being effective, while simpler classifiers may fail to capture complex phishing patterns. [7], [8].

To address these limitations, ensemble learning techniques have been increasingly adopted to improve detection accuracy and robustness [10], [11]. Comparative studies evaluating multiple classifiers have shown that Extreme Gradient Boosting (XGBoost) consistently outperforms traditional

models such as Logistic Regression and Random Forest, particularly in handling non-linear feature interactions and class imbalance [10], [11]. Additional research focusing specifically on XGBoost-based phishing detection further confirms its scalability, efficiency, and suitability for practical phishing detection systems [12].

Various deep learning methods, which can learn complex feature representations, have also been explored for phishing detection. Convolutional Neural Network-based models have demonstrated high detection accuracy by capturing hierarchical feature patterns [9]. However, these approaches often require large training datasets and incur high computational overhead and latency, which limits their applicability in lightweight and real-time phishing detection environments [3], [9].

More recently, studies also show that relying solely on URL-based machine learning is insufficient to address the complexity of modern phishing attacks [5], [6]. Hybrid phishing detection frameworks that integrate machine learning with trust-based indicators, such as SSL certificate metadata and DNS-related features, have been shown to improve robustness and reduce false-positive rates [5], [13], [14]. The trust-based indicators supplement the contextual information to machine learning-based predictions, especially for newly registered or suspicious domains, and help in developing practical hybrid phishing detection systems.

3. PROBLEM DEFINITION & OBJECTIVES

Phishing is one of the fastest-growing cyber-attacks, in which attackers use spoofed URLs to create fraudulent websites that imitate legitimate platforms in order to deceive users into revealing sensitive information such as login credentials, passwords, and financial details. With the rapid growth of internet usage and increased reliance on web-based applications, users have become more vulnerable to phishing attacks that exploit trust, familiarity, and a lack of technical awareness [1].

In response to this threat, several phishing-detection websites and online tools have been developed. However, most existing systems rely primarily on static blacklists, rule-based techniques, or analysis of a limited number of URL features. Such approaches are largely reactive and ineffective against newly generated, short-lived, or zero-day phishing URLs, leading to low detection accuracy and reduced reliability in real-time environments [1], [4]. These limitations significantly increase the risk of false negatives and false positives, thereby reducing user trust in existing solutions.

To address these challenges, this research proposes the development of an intelligent phishing-detection website that allows users to submit a URL and receive an instant classification indicating whether the link is phishing or legitimate. The proposed system integrates machine learning-based URL classification with external threat intelligence sources to improve detection accuracy beyond that of existing solutions.

By combining automated learning with trust-based indicators, the framework aims to provide a more adaptive, accurate, and reliable phishing-detection mechanism suitable for real-time deployment [2], [5].

The core objective of the study is to design and develop an intelligent phishing-detection website that can classify a user-submitted URL as either phishing or legitimate in real time with high accuracy. The proposed study attempts to go beyond the efficacy achieved by state-of-the-art phishing-detection platforms, whose failures are attributed to the shortcomings of the conventional blacklist-based and rule based approaches [1], [4].

A secondary objective is to analyse and extract pertinent lexical and structural features from URLs that effectively differentiate phishing sites from legitimate ones. These features are used in training a machine-learning-based classification model that can identify malicious URLs, including newly generated and previously unseen phishing links [2], [4].

This study will also include the incorporation of threat-intelligence sources from outside, such as DNS reputation analysis and SSL certificate trust evaluation, to enhance the reliability and robustness of the detection framework [13], [14]. Finally, the proposed system, by hybridizing machine-learning predictions with trust-based indicators via a hybrid decision strategy, aims at a reduction of both false positives and false negatives, thus granting an adaptive, more accurate, and dependable mechanism of phishing detection suitable for real-time deployment [5], [15].

4. PROPOSED METHODOLOGY

4.1 URL Feature Representation

URL feature representation is a major component of the proposed phishing-detection framework, since phishing URLs often contain peculiar lexical and structural features [1], [4]. During this stage, the input URL is represented as a numeric feature vector by extracting features related to URL length, special character count, occurrence of IP addresses, number of subdomains, occurrence of suspicious keywords, type of protocol used, and depth of URL path [2], [4]. These features capture abnormal URL patterns that are related to phishing activities [4]. Further, after normalization and encoding, the feature vector will be presented as an input to the machine learning-based classification module [2], [5].

4.2 ML-based Classification of URLs

The proposed framework uses the Extreme Gradient Boosting algorithm to classify URLs as this technique is accurate and quick in processing structured feature data [10], [11]. XGBoost is an ensemble learning technique that involves several decision trees in succession and each of these trees learns from the errors of the previously developed tree [12].

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i)$$

where y_i denotes the predicted label for the input feature vector x_i , K represents the total number of decision trees in the model, and f_k corresponds to the prediction function of the k^{th} decision tree. The training objective of the model is to minimize the following loss function:

$$\mathcal{L} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

It predicts the class of the URL with a function involving a loss term and a regularization term to prevent overfitting [11], [12]. The model returns a probability score when it is used on a URL, showing how likely it is that the provided URL is either phishing or legitimate [10].

4.3 DNS Reputation Analysis Using APIs

DNS reputation analysis is also built into the proposed framework to appraise a URL's domain for trust using external threat-intelligence APIs. The following are some of the domain-related attributes to be looked at: domain age, registration details, history of resolution behaviour, and any known malicious activity. Phishing sites usually have newly registered or short-lived domains, which makes DNS-based indicators very effective for early-stage detection [14]. The risk score of the domain derived by querying the reputation API supplements machine learning-based URL classification in enhancing the overall robustness and accuracy of phishing detection [5], [14].

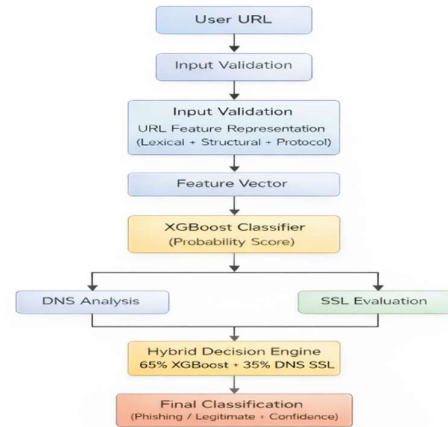


Fig.4.1. Architecture of the Proposed Hybrid Phishing Detection System

4.4 SSL certificate trust evaluation using APIs

The trust analysis of the SSL certificates using the APIs also enables the assessment of the trustworthiness of a website by considering the SSL/TLS certificate information using trusted security components [13]. This includes the validation of the validity, issuer, age of the certificate, if the certificate has expired, and the level of encryption used [13]. Since phishing websites normally use an invalid, self signed, and newly issued certificate to impersonate the original website, the analysis of the SSL trust, therefore, becomes an essential indicator of malicious activity [13]. Incorporating the SSL trust analysis in the proposed approach enhances the trustworthiness of the detection technique [5], [13].

4.5 Hybrid Decision Strategy

The hybrid decision approach makes use of the results obtained from both machine learning-based classification of URLs as well as DNS analysis and SSL trust evaluation to come up with a conclusive decision on the reliability of the URL. As indicated above, using single classifiers for phishing detection might result in incorrect classification results. Therefore, using multiple complementary features for classification purposes can result in improved results [1], [5]. In this work, to achieve improved adaptability for anti-phishing applications, the result from the XGBoost classifier is validated using trust level information obtained from DNS/SSL analysis. This two-layer approach ensures improved resistance to zero-day phishing attacks as well as reduced instances of false alarms [5], [14], [15].

5. RESULTS AND PERFORMANCE

5.1 URL Detection Model Results and Performance

The proposed hybrid phishing detection framework was evaluated using standard classification metrics to assess its effectiveness in distinguishing phishing URLs from legitimate ones. The XGBoost-based URL detection model produced the following outcomes:

-Overall Accuracy: 99%

-Precision: 0.99

-Recall: 0.99

-F1-Score: 0.99

Confusion Matrix:

The confusion matrix illustrates the classification performance of the proposed model by presenting the distribution of true positives, true negatives, false positives, and false negatives across both classes.

The confusion matrix shows that 6512 legitimate URLs and 6523 phishing URLs were correctly classified, with only a very small number of misclassifications, demonstrating the robustness and reliability of the proposed model.

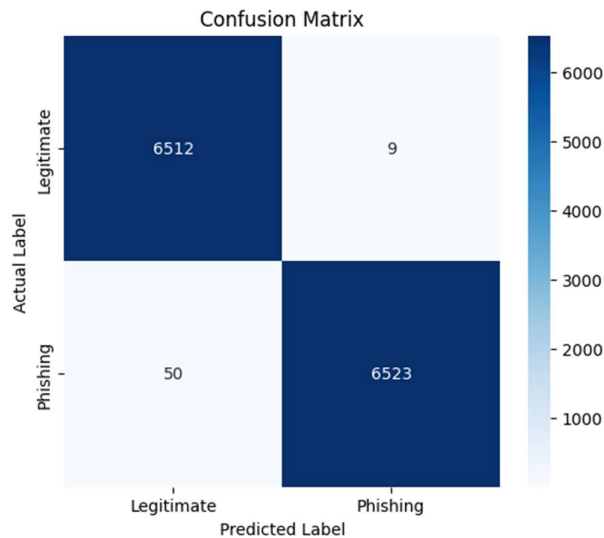


Fig. 5.1. Confusion Matrix for XGBoost-based URL Detection

-Classification Report:

Class	Precision	Recall	F1-Score	Support
Bad	0.99	1.00	0.99	3314
Good	1.00	0.99	0.99	3210
Accuracy	—	—	0.99	—
Macro Avg	0.99	0.99	0.99	6524
Weighted Avg	0.99	0.99	0.99	6524

Table 5.1 Classification Report for URL Detection Model

The classification report confirms consistently high precision, recall, and F1-score values for both phishing and legitimate classes, indicating balanced and reliable performance.

5.2 Performance Evaluation & Interpretation

-High Detection Accuracy:

The proposed model achieved an overall accuracy of 99%, indicating its strong capability in learning discriminative URL patterns associated with phishing attacks.

-Effective Generalization:

The consistent performance across training and testing datasets demonstrates that the model generalizes well to unseen URLs, reducing the risk of overfitting.

-Insights from Confusion Matrix:

The confusion matrix reveals extremely low false-positive and false-negative rates, which is crucial for phishing detection systems where misclassification can lead to serious security risks.

-ROC Curve Analysis:

The Receiver Operating Characteristic (ROC) curve achieved an Area Under the Curve (AUC) value of 1.00, highlighting excellent class separability.

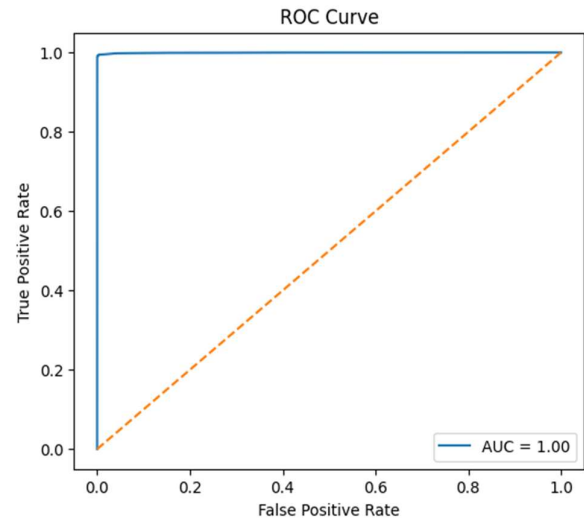


Fig. 5.2. ROC Curve for URL Classification

5.3 Feature Importance and Model Stability Analysis

-Precision-Recall Curve Analysis:

The Precision-Recall curve maintains high precision across almost all recall values, demonstrating the model's ability to minimize missed phishing detections.

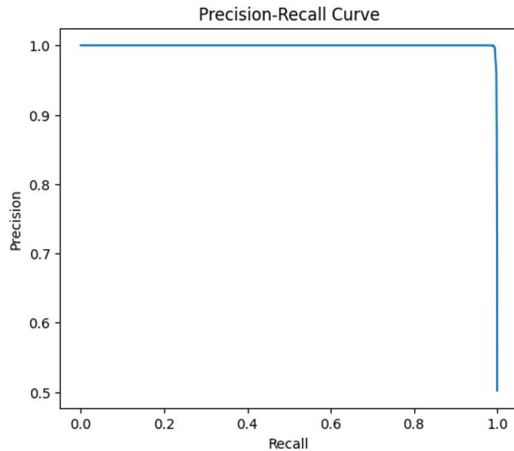


Fig. 5.3 Precision-Recall Curve for URL Classification

-Feature Importance:

Feature importance analysis indicates that URL path length, slash count, and HTTPS usage are the most influential features contributing to phishing detection. This aligns with known phishing strategies where attackers exploit complex URL structures.

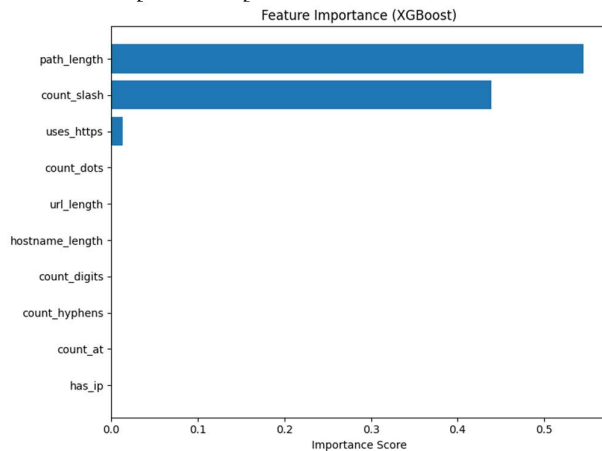


Fig. 5.4. Feature Importance Analysis using XGBoost

-Training vs Testing Performance:

The comparison between training and testing precision, recall, and F1-score confirms consistent model behavior, validating its stability and robustness.



Fig. 5.5 Training vs Testing Performance Metrics

6. LIMITATIONS & FUTURE WORK

6.1 LIMITATIONS

Although the proposed phishing detection system benefits from a hybrid decision strategy that combines machine learning-based URL analysis with DNS and SSL certificate verification, several limitations remain.

First, the framework assigns a higher decision weight to the XGBoost classifier, which contributes approximately 65% of the final prediction. While this improves detection accuracy for previously observed phishing patterns, it also increases sensitivity to the quality and diversity of the training dataset. If the dataset does not adequately represent newly emerging or obfuscated phishing techniques, the model may produce inaccurate predictions until retrained with updated samples.

Second, the remaining 35% contribution from DNS and SSL-based indicators introduces dependencies on external domain-level information. Although these signals enhance trust validation, they are not always reliable. Many phishing websites now deploy valid SSL certificates issued by trusted certificate authorities or leverage compromised legitimate domains. In such cases, SSL and DNS features may incorrectly suggest legitimacy, thereby reducing their effectiveness within the hybrid decision process.

Another limitation arises from the fixed weighting scheme used to combine machine learning predictions with DNS and SSL analysis. The static contribution ratio (65% ML and 35% DNS/SSL) does not adapt to changing threat contexts. For example, in scenarios where reputation data is sparse or misleading, the assigned weight may not reflect the true reliability of DNS and SSL signals, potentially affecting overall detection performance.

In addition, the framework primarily focuses on URL-level characteristics and domain reputation, without incorporating webpage content, visual similarity, or user interaction behavior. As a result, phishing websites that closely imitate legitimate pages while maintaining seemingly trustworthy URLs and certificates may evade detection.

Finally, the system evaluates URLs in isolation and does not consider session-level or behavioral context, such as redirection chains or temporal browsing patterns. This limitation may reduce resilience against multi-stage phishing attacks that rely on complex navigation flows to deceive users.

6.2 FUTURE WORK

Future research can focus on enhancing the adaptability and robustness of the proposed hybrid phishing detection framework. One important direction involves replacing the current static weighting scheme, where machine learning contributes 65% and DNS/SSL

indicators contribute 35%, with a dynamic or context-aware weighting mechanism. Such an approach could adjust the influence of each component based on factors such as domain age, certificate validity patterns, or confidence scores produced by the classifier.

Another promising extension is the integration of webpage content and visual feature analysis. Incorporating deep learning models, such as Convolutional Neural Networks (CNNs), to analyze webpage screenshots, HTML structures, and textual similarities would strengthen detection capabilities against phishing websites that use legitimate-looking URLs and valid SSL certificates. This addition would complement the existing URL-level and reputation-based features.

The framework can also be improved through the adoption of adaptive or incremental learning techniques. Allowing the XGBoost model to update continuously using newly identified phishing samples would reduce performance degradation caused by evolving attack strategies and improve long-term detection accuracy.

From a deployment perspective, extending the system to browser extensions and mobile platforms would enable real-time phishing prevention directly within user environments. Such integration would improve usability and allow immediate detection during user interaction with potentially malicious websites.

Finally, future work may incorporate behavioral and contextual analysis, including redirection chains, session-level activity, and temporal access patterns. In addition, integrating explainable artificial intelligence (XAI) techniques could provide interpretable explanations for hybrid decisions, helping users and security analysts understand how machine learning predictions and DNS/SSL signals jointly influence final outcomes.

7. CONCLUSION

Phishing attacks continue to represent a persistent and evolving cybersecurity threat, driven by attackers' ability to rapidly modify techniques and exploit trust mechanisms. Traditional detection methods, such as static blacklists and rule-based systems, often fail to respond effectively to newly generated phishing websites and short-lived attack campaigns.

This work, presented as hybrid phishing detection framework that combines machine learning-based URL analysis with DNS and SSL certificate reputation assessment. The detection task was formulated as a supervised binary classification problem, where an XGBoost model serves as the primary decision component, contributing approximately 65% of the final prediction. The remaining 35% is derived from DNS resolution and SSL certificate analysis, which

provide complementary domain-level trust validation. This hybrid design enables a balanced detection strategy that leverages both data-driven learning and real-world reputation signals.

The proposed system was implemented using a real-time backend architecture and a web-based frontend, facilitating practical and accessible phishing detection. Experimental evaluation demonstrated reliable detection performance, with high accuracy and controlled false-positive rates. The weighted hybrid approach effectively mitigates weaknesses associated with relying solely on machine learning or reputation-based methods.

Overall, PhishX offers a scalable, efficient, and cost-effective phishing detection solution that operates without dependence on proprietary or paid threat intelligence services. Although certain limitations remain—such as static weighting and the absence of content-level analysis—the framework provides a solid foundation for future enhancements. This research contributes to phishing detection literature by demonstrating the feasibility and effectiveness of a weighted hybrid approach for real-world cybersecurity applications.

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